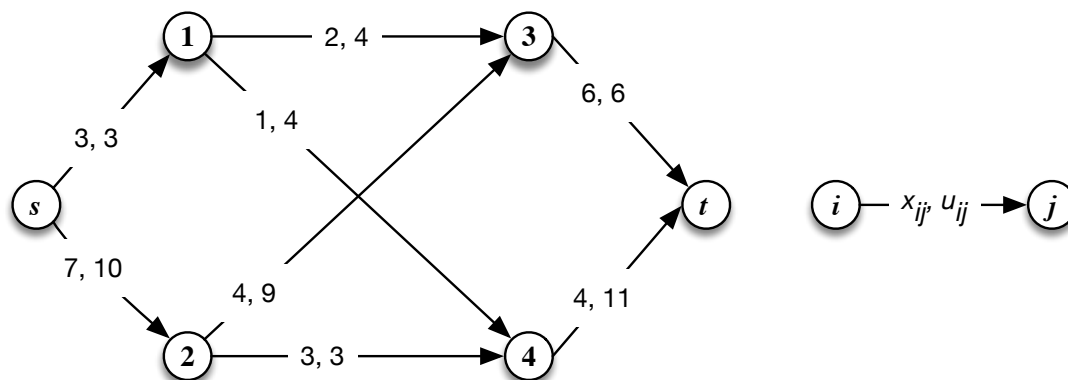


First part : Closed book

Total score: 21 pts. Minimum score to pass: 11 pts. Tentative scores: 1) 4 pts, 2) 5 pts, 3) 5 pts, 4) 3 pts. 5) 4 pts.

1. Determine the maximum flow from s to t in the following graph, **starting from the given flow**. Illustrate the intermediate steps and report the minimum capacity cut. Does the solution change (maximum flow or minimum cut) if the capacity of arc $(1, 4)$ increases of one unit? Justify your answer.



2. Consider the following linear programming problem:

$$\begin{aligned}
 \min \quad & -5x_1 + 7x_2 + 10x_3 + 10x_4 \\
 & -x_1 + x_2 - 2x_3 + 2x_4 \geq 3 \\
 & -x_1 + 3x_3 - x_4 \geq 3 \\
 & x_1, x_2, x_3, x_4 \geq 0
 \end{aligned}$$

Write the dual and the complementary slackness equations. For each of the following solutions $x' = [0, 0, 9/4, 15/4]$, $x'' = [0, 5, 1, 0]$ derive the complementary dual solutions and discuss the primal and dual feasibility.

3. Consider the following Integer Linear Programming problem:

$$\begin{aligned}
 \max \quad & x_1 - 5x_2 \\
 & 4x_1 - 12x_2 \leq 15 \\
 & 4x_1 + 12x_2 \leq 23 \\
 & x_1, x_2 \in \mathbb{Z}^+
 \end{aligned}$$

Add the slack variables (x_3 and x_4) to the first and the second constraints, and consider the optimal base $B = \{1, 4\}$ of the LP relaxation. Generate the Gomory cuts associated with the fractional components of the solution. Justify your answer and give the geometric representation of the feasible region and of the cut(s).

4. Consider the following linear programming problem $\min\{cx : Ax \geq b, x \geq 0, x \in \mathcal{R}^n\}$, where A is an $m \times n$ matrix, b an m component column vector and c an n component row vector. Write the dual. Given two solutions $\bar{x}$ and $\bar{y}$, feasible for the primal and the dual, respectively, prove that $c\bar{x} \geq \bar{y}b$.

APML question in the rear

Write the definition in the ampl modelling language of the following:

- a set I .
- a set J of all the integers between 1 and 10.
- a positive variable x_{ij} with $i \in I$ and $j \in I$.
- a parameter b_j with $j \in J$.

Then using the defined variables and parameters write in the AMPL modelling language an objective function that maximizes the following quantity:

$$\max \sum_{i \in I} \sum_{j \in J: i \neq j} b_j x_{ij}$$

ANSWER HERE